

SOUMIK SARKER ANIK

 LinkedIn |  soumikportfolio.vercel.app |  ssanik@mun.ca |  709 853 4521 |  St John's, NL

SUMMARY

Master's graduate in Software Engineering with **three** years of research and industry experience in the field of machine learning and data science. Highly proficient in deep learning models and fuzzy logic systems. Co-author of three journal articles and two conference papers with a focus on predictive modeling and the analysis of complex geospatial, environmental, healthcare, and time series datasets. Fast learner, self-motivated, and tech enthusiast with excellent analytical and coding skills proven through work experience.

EDUCATION

Memorial University of Newfoundland	2023 – 2025
Masters of Applied Science, Software Engineering	St John's, NL
North South University	2017 – 2021
Bachelor of Science, Computer Science – GPA 3.77/4.00 (Magna Cum Laude)	Dhaka, Bangladesh

WORK EXPERIENCE

Junior AI Engineer	April 2026 – Current
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C-Core

- Building risk-scoring detectors that flag vessels of interest from AIS behaviour.
- Developing models that correlate vessel targets across satellite imagery, AIS, and RF data for maritime domain awareness.
- Building production ML pipelines covering training data preparation, model evaluation, and monitoring.

Software/Research Developer (Intern)	Sep 2024 – Dec 2024
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Memorial University

- Developed and implemented unsupervised ML models for anomaly detection in industrial machinery time series data.
- Performed necessary feature engineering and preprocessing steps on the data to enhance model performance.
- Utilized Azure Databricks and Azure Data Explorer (ADX) for data extraction and contributed to designing the solution architecture.

University Research Assistant	Sep 2019 – Feb 2022
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North South University

- Published three journal articles and two book chapters showcasing expertise in advanced ML algorithms across domains, with over 80 citations.
- Implemented different classification, clustering, and hybrid neural network models with my research team of five people.
- Conducted research in diverse domains, including healthcare analytics (COVID-19 transmission modeling), geospatial analysis (flood risk assessment), and transportation safety (accident severity prediction).

PROJECTS AND PUBLICATIONS

- **Flood Risk Assessment in Bangladesh** ([Publication](#) – [Geocarto International, 2022](#))
GIS-based national-scale flood risk modeling using hybrid Deep Neural Network (DNN) and fuzzy AHP methods.
- **Hybridized SVR for Flood Susceptibility Mapping** ([Publication](#) – [IEA/AIE 2021](#))
Applied Genetic Algorithm-optimized Regression models for flood susceptibility prediction using GIS data.
- **Road Crash Severity Prediction Using Machine Learning** ([Publication](#) – [ICCCI 2020](#))
Machine learning-based approach using Cluster Analysis, Random Forest, and C5.0 models for crash prediction.
- **Time Series Anomaly Detection in Gas Compression Data** ([Project](#))
Developed unsupervised models (DBSCAN, Isolation Forest) to identify equipment anomalies in sensor data.

SKILLS

- **Programming Languages:** Python, R, SQL, C/C++, MATLAB, GIS Softwares (ArcGIS)
- **MLOps and DevOps:** CI/CD pipelines, MLflow, Git, Azure Databricks, Azure Data Explorer (ADX)
- **Machine Learning and NLP:** scikit-learn, Keras, TensorFlow, PyTorch, NLTK, NumPy, pandas
- **Domain Specific:** Healthcare Data Analytics, Computer Vision, Hybrid ML models, Deep Learning Models, Predictive Modelling, Anomaly Detection
- **Data Visualization:** Matplotlib, Seaborn, Power BI